

TIAN-AO (TEO) REN

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RESEARCH INTEREST

My research lies at the intersection of robotics, haptics, and artificial intelligence for intelligent physical interaction in medicine. I am interested in developing multimodal tactile sensing systems, ultrasound-compatible interfaces, and AI-driven contact modeling frameworks that enable robots to perceive, model, and augment expert human touch. More broadly, I seek to bridge hardware innovation, computational learning, and clinical insight to advance human-robot collaboration, skill transfer, and intelligent robotic assistance in healthcare.

EDUCATION

Ph.D. Candidate in Mechanical Engineering (Robotics & Design) Jan 2025 - **Present**
Stanford University California, U.S.

- Advisor: Prof. Mark Cutkosky, PI of [Biomimetics and Dextrous Manipulation Lab\(BDML\)](#)
- **Stanford Robotics Scholars Fellowship**, Inaugural Recipient (Co-Advised by Prof. Oussama Khatib)

M.S. in Mechanical Engineering (Design Methodology) Sep 2023 - Jun 2025
Stanford University California, U.S.

- Advisors: Prof. Allison Okamura, PI of [Collaborative Haptics and Robotics in Medicine Lab\(CharmLab\)](#)
- Prof. Mark Cutkosky, PI of [Biomimetics and Dextrous Manipulation Lab\(BDML\)](#)

B.Eng. in Robotics Engineering Sep 2019 - Jun 2023
Beijing University of Chemical Technology Beijing, China

PUBLICATION *INDICATE EQUAL CONTRIBUTION

Journal Articles

- J. Lai*, **T.-A. Ren***, P. Ye, J. Sun, and H. Ren, "Gravity-Aware Proactive Joint-Level Compensation for Portable Soft Slender Robots Using a Single IMU and Real-Time Simulation," *International Journal of Robotics Research (IJRR)*, 2026. [\[DOI\]](#)
- E. Rennich, **T.-A. Ren**, J. Kim, J. Di, T. Chen, and M. Cutkosky, "Gentle Grasping with Gecko-Inspired Adhesives in Extreme Environments," *IEEE Transactions on Field Robotics (T-FR)*, 2025. [\[DOI\]](#)
- J. Lai*, Y. Liu*, **T.-A. Ren**, Y. Ma, T. Zhang, J. Y. Teoh, M. Cutkosky, and H. Ren, "Single Twistable Tendon-Driven Continuum Robots," *Nature Communications*, 2025.
- J. Lai*, **T.-A. Ren***, W. Yue, S. Su, J. Y. K. Chan, and H. Ren, "Sim-to-Real Transfer of Soft Robotic Navigation Strategies That Learn from Virtual Eye-in-Hand Vision," *IEEE Transactions on Industrial Informatics (T-II)*, 2023. [\[DOI\]](#) [\[Video\]](#)
- G. Wang, **T.-A. Ren**, J. Lai, L. Bai, and H. Ren, "Domain Adaptive Sim-to-Real Segmentation of Oropharyngeal Organs," *Medical & Biological Engineering & Computing (MBEC)*, 2023. [\[DOI\]](#)

Conference Papers

- **T.-A. Ren**, J. Garcia, S. Hong, J. Grinberg, H. Choi, J. Di, H. Li, D. Grinberg, and M. Cutkosky, "Multimodal Sensing for Robot-Assisted Sub-Tissue Feature Detection in Physiotherapy Palpation," *Design of Medical Device Conference (DMD)*, 2026. [\[DOI\]](#) [\[Oral\]](#)
- J. Di, K. Hoffmann, T. Chen, **T.-A. Ren**, and M. Cutkosky, "SLAP: Slapband-Based Autonomous Perching Drone with Failure Recovery for Vertical Tree Trunks," *IEEE Aerospace Conference*, 2026. [\[DOI\]](#)
- C. Ng, H. Gao, **T.-A. Ren**, J. Lai, and H. Ren, "Contact-Aided Navigation of Flexible Robotic Endoscope Using Deep Reinforcement Learning in Dynamic Stomach," *IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2025. [\[DOI\]](#) [\[Best Paper Award, Oral\]](#)
- C. Xing, H. Li, Y. Wei, **T.-A. Ren**, T. Tu, Y. Lin, E. Schumann, W. Zheng, and M. Cutkosky, "TacCap: A Wearable FBG-Based Tactile Sensor for Seamless Human-to-Robot Skill Transfer," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. [\[DOI\]](#)

- **T.-A. Ren**, W. Liu, T. Zhang, H. Ren, and J. Lai, “Three-Dimensional Morphological Reconstruction of Millimeter-Scale Soft Continuum Robots Based on Dual Stereo Vision,” *IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2024. [\[DOI\]](#) [\[Oral\]](#)
- C. Ng, H. Gao, **T.-A. Ren**, J. Lai, and H. Ren, “Navigation of Tendon-Driven Flexible Robotic Endoscope Through Deep Reinforcement Learning,” *IEEE International Conference on Advanced Robotics and Its Social Impacts (ARSO)*, 2024. [\[DOI\]](#)

Posters & Demonstrations

- **T.-A. Ren**, S. Hong, J. Garcia, J. Grinberg, and M. Cutkosky, “PhysioVisionFT: A Multimodal Sensor Framework for Sub-Tissue Feature Detection in Physical Therapy,” Bay Area Robotics Symposium, 2025.
- **T.-A. Ren** and M. Cutkosky, “Stickybot IV,” Stanford Robotics Center Launch Symposium. [\[Demo\]](#)
- **T.-A. Ren**, Z. Yu, B. Vuong, A. Meola, and A. Okamura, “Concentric Agonist-Antagonist Robots for Neurosurgical Applications,” *IEEE RoboSoft Workshop*, 2024. [\[Best Submission for Lightning Talk\]](#) [\[PDF\]](#)
- G. Wang, **T.-A. Ren**, J. Lai, L. Bai, and H. Ren, “Sim-to-Real Segmentation in Robot-Assisted Transoral Tracheal Intubation,” *IEEE ICRA Workshop*, 2023. [\[DOI\]](#) [\[Best Poster Award\]](#)

SERVICES

- **Journal Reviewer**: IEEE Transactions on Medical Robotics and Bionics (**T-MRB**); IEEE Transactions on Industrial Informatics (**T-II**); Machine Intelligence Research (**MIR**).
- **Conference Reviewer**: IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024, 2026; IEEE International Conference on Robotics and Automation (**ICRA**), 2024–2026.

TEACHING & MENTORSHIP EXPERIENCES

- **TA for ME 327: Design and Control of Haptic Systems** (Lecturer: Prof. Allison Okamura) Spring 2025
- **Mentee**: Jorge Garcia (M.S. at Stanford)

HONORS & AWARDS

- **Bio-X Travel Award** at Stanford University 2026
- **Outstanding Graduates in Beijing** 2023
- **National Scholarship of China** 2022
- **Best Paper Award** in *The 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO 2025)*
- **Best Poster Award** in *The 2023 IEEE International Conference on Robotics & Automation Workshop (ICRA 2023)*